Mapping Wildfire Vulnerability in Colorado

Machine Learning and Satellite Embeddings



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Why?



1. Wildfire frequency and intensity are increasing in Colorado
2. Fires are getting closer and closer to urban areas
3. Traditional fire risk maps are coarse and often don't use satellite imagery
4. Better predictions improves preparedness and mitigation

Study Area & Data

- **Study region**: CO State Boundary TIGER/2018/States
  - Used to clip predictor layers
- **Elevation**: USGS/SRTMGL1_003 (30m)
  - Elevation, slope, aspect
- **Vegetation and Fuel Variables**
  - Tree Cover: UMD/hansen/global_forest_change_2022_v1_10
  - NDVI: MODIS/061/MOD13A2
  - Land Cover: MODIS/061/MCD12Q1
    - LC_Type1
- **AlphaEarth/Satellite Embeddings**
  - GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL
  - Encodes texture, land pattern, and spectral features
- **Fire Labels** (Training Target): USFS/GTAC/MTBS
  - Rasterized to produce 1/0 mask

Sample Region:

- 5000m buffer around burned area
- 1000 burned
- 2000 unbured
- Used only for extracting training samples
- Helped efficiency/memory
- More balanced data

```
sample_region = mtbs.geometry().bounds().buffer(5000)
```

Preprocessing Steps

- Define study region (Colorado)
- Load predictor layers from GEE
- Stack predictors into multiband image
- Create fire labels by rasterizing MTBS burned area polygons
- Create simplified sampling region (MTBS bounds + Buffer)
    - Reduced memory errors
    - Meaningful fire vs non fire comparison
- Add labels to predictors (burned mask as band)
- Stratified sampling to get more balanced burned vs. unburned pixels
    - Burned pixels are rare, prevent unburned bias
- Create DataFrame and drop missing values

Models

- **Logistic Regression**
  - Baseline classification
  - Can linear boundaries separate fire vs. non-fire
- **Random Forest**
  - Strong baseline for spatial/vegetation features
  - Good for feature importance
- **Gradient Boosting**
  - Captures more subtle interactions than RF
- **XGBoost**
  - Higher performance classifier
  - Works well with tabular and remote sensed data, and complex interactions between features
- **SVM**
  - Captures complex, non-linear boundaries
  - Uses probability = True for AUC, gives probability and not just label
  - 75/25 split
  - Hinge loss function
    - Good for the fine non-linear boundary between fire/non-fire vulnerability areas
    - Penalizes more near the boundary
    - Effective for high dimensional embeddings

**RandomizedSearchCV used to optimize the model, tuned for ROC-AUC

```
search = RandomizedSearchCV(
    estimator=model,
    param_distributions=grid,
    n_iter=20,
    scoring="roc_auc",
    cv=3,
    random_state=42,
    n_jobs=-1
)
```

```
param_grids = {
    "LogReg": {
        "C": np.logspace(-3, 3, 20),
        "penalty": ["l1", "l2"],
        "solver": ["liblinear"]
    },

    "RandomForest": {
        "n_estimators": [100, 200, 300, 500],
        "max_depth": [None, 5, 10, 15, 20],
        "min_samples_split": [2, 5, 10],
        "min_samples_leaf": [1, 2, 4],
    },

    "GradientBoosting": {
        "n_estimators": [100, 200, 300],
        "learning_rate": [0.01, 0.05, 0.1, 0.2],
        "max_depth": [2, 3, 4, 5],
        "subsample": [0.6, 0.8, 1.0]
    },

    "SVM": {
        "C": [0.1, 1, 10, 50, 100],
        "gamma": ["scale", "auto", 0.001, 0.01, 0.1, 1],
        "kernel": ["rbf"]
    }
```

```
param_grids["XGBoost"] = {
    "n_estimators": [200, 300, 500],
    "learning_rate": [0.01, 0.05, 0.1],
    "max_depth": [3, 4, 5, 6],
    "subsample": [0.7, 0.8, 1.0],
    "colsample_bytree": [0.7, 0.8, 1.0]
}
```

Model Results

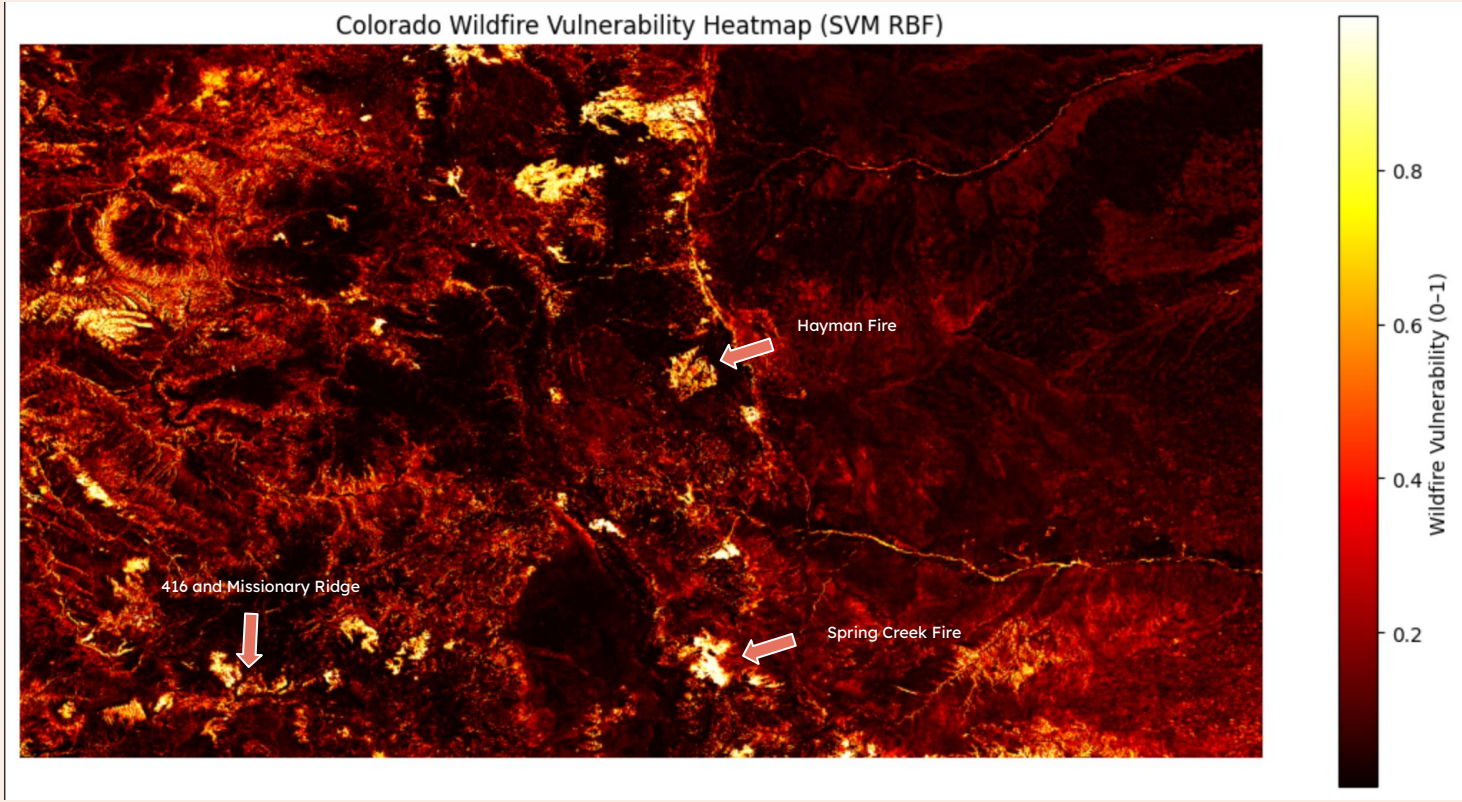
Model	Accuracy	ROC_AUC	F1
SVM	0.896552	0.904375	0.600000
RandomForest	0.862069	0.835000	0.000000
LogReg	0.905172	0.830000	0.521739
XGBoost	0.913793	0.806250	0.583333
GradientBoosting	0.913793	0.780625	0.545455

Why ROC-AUC

- Dataset is imbalanced (unburned > burned pixels)
- Accuracy and F1 can be misleading
- Evaluates model via probability, not single cutoff
- Good for continuous map

Results Cont.

Model	Accuracy	ROC-AUC
Embeddings Only	0.844000	0.878116
ENV Only	0.742667	0.744556



Model Predicted Wildfire Vulnerability

- Lighter colors = higher probabilities that environmental conditions resemble historically burned areas

Insights

Spatial Patterns

- At risk zones cluster along forested foothills at mid elevation
- Predicted hotspots align with historical MTBS burn scars
- Low risk appears in irrigated valleys, agricultural plains and high alpine tundra
- Transition zones are consistently at risk

ENV Drivers

- Low RH means dry fuels resulting in higher probability of ignition
- South facing slopes increase risk because of faster drying
- NDVI and fuel type help identify high-fuel areas
- Distance to roads and nighttime lights showed us ignition likelihood near WUI

Emb vs Env

- ENV variables explain the direct physical mechanisms of fire behavior
- Sat Embs improved the models ability to recognize broader spatial context
- Embeddings produced strong results on their own
- Embeddings helped in areas with mixed or complex vegetation

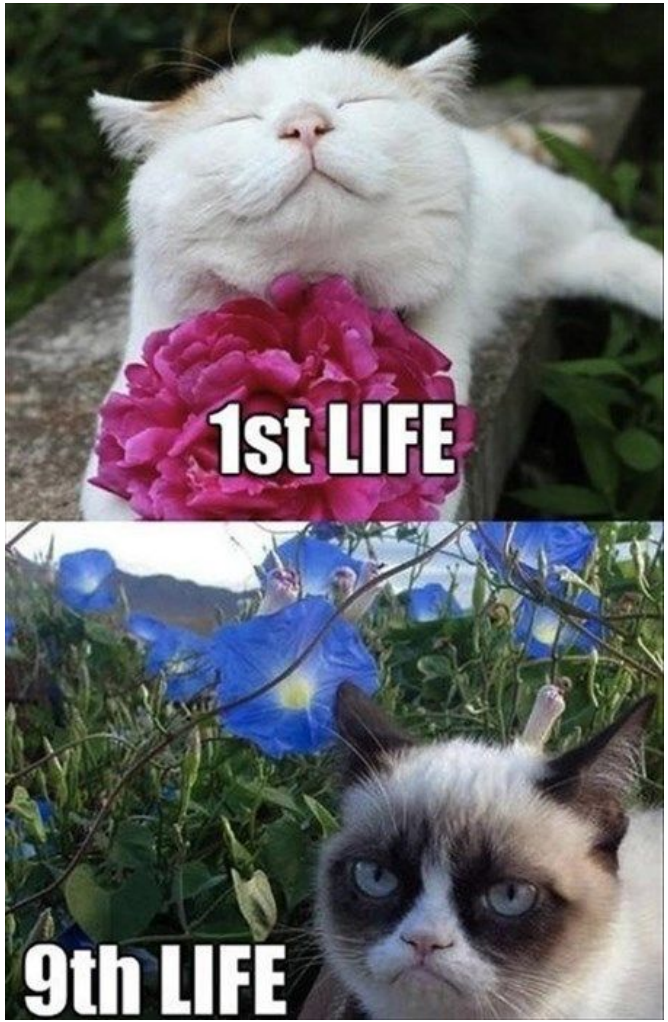
HR vs LR zones

- HR zones overlap with dense conifer populations
- LR zones are often in irrigated ag regions or high elevation tundra
- Medium risk zones frequently appear along the WUI

	feature	importance_mean	importance_std
21	treecover	0.096552	0.042722
0	A00	0.077586	0.028330
5	A05	0.036207	0.014832
15	A15	0.036207	0.014832
14	A14	0.032759	0.016716
6	A06	0.018966	0.013793
18	landcover	0.017241	0.012191
7	A07	0.015517	0.016716
4	A04	0.015517	0.010053
13	A13	0.013793	0.025919
20	slope	0.006897	0.012670
19	ndvi	0.005172	0.010345
3	A03	0.005172	0.011694
11	A11	0.001724	0.025222
16	aspect	0.001724	0.006451
12	A12	-0.001724	0.016716
2	A02	-0.003448	0.018570
10	A10	-0.005172	0.012902
1	A01	-0.006897	0.012670
17	elev	-0.008621	0.019658
9	A09	-0.012069	0.015989
8	A08	-0.020690	0.006897

Limitations

- **Imbalanced data**
  - fire data is inherently imbalanced
    - Non-burned > burned
  - Our dataset was 2:1 ratio of non-fire to fire samples
- **GEE data limitations**
  - Download limit on data
  - Limited to coarse resolution
- **Static Data**
  - Used Annual and long term composites
  - No seasonality
  - Problem - Some fires happen from short-term weather extremes aren't taken into consideration in our model
- **Predictor Proxy Variables**
  - Some predictors like Nighttime lights and population density are proxy variables
  - They approximate conditions rather than measuring true physical drivers



Conclusion

- Satellite embeddings were very important and useful predictors
- ENV variables helped represent physical drivers of wildfire conditions
- SVM was our best model
- Despite limitations, the model still produced clear and interpretable spatial patterns. .
- Predicted vulnerability patterns align visually with known high risk areas in Colorado

Future Work

- **Incorporate More Predictors**
  - **Fuel Moisture indices, other climate variables**
- **Improve sampling to wider region**
- **Temporal Dynamics**
- **Higher Resolution**